

Vaibhav Saini

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Summary

I'm an early career AI-focused Software Engineer who believes the best systems are fast, resilient, and secure; but I never stop thinking about whether something feels intuitive to the person on the other end. I'm a curious, quick learner who picks up new tools fast and goes deep when it matters, caring about latency, failure modes, and whether AI output can actually be trusted. I collaborate well across in person and remote teams, working efficiently no matter the setup. On OPT, looking for a full-time role.

Education

Master of Science in Information Technology (Security)

May 2026

Arizona State University, Tempe, AZ

GPA: 3.97 / 4.0

Bachelor of Engineering in Computer Science (Data Science)

June 2023

MMDU, Ambala, Haryana, India

GPA: 7.62 / 10.0

Experience

Software Engineer | Badoliasoft

Oct 2023 - Jun 2024

- Implemented JWT auth with HMACSHA512 hashing, 3 tier RBAC, and token backlisting, securing a multi-tenant fintech platform for 100+ users.
- Built real time notification pipeline using SignalR with RabbitMQ fanout exchange fallback; integrated Azure Blob storage with CDN for file delivery.
- Configured IP-based rate limiting, global exception middleware with structured logging, and Redis distributed caching to reduce DB load and improve reliability.
- Contributed to a full-stack Angular + .NET 8 platform (200+ C# files, 10 REST controllers) spanning user management, messaging, and project workflows.

Technical Skills

- **Language & Frameworks:** Python, Java, FastAPI, TypeScript, Node.js, React, .NET, Angular
- **AI & Machine Learning:** LLMs, RAG, Embeddings, Sentence-Transformers, LLM Fine-Tuning, Prompt Engineering, Vector Search, LangChain
- **Cloud & DevOps:** AWS, Azure, Docker, CI/CD
- **Databases:** PostgreSQL, DynamoDB, Pinecone, ChromaDB, SQL Server, Redis, Vector Database
- **Security:** Splunk, Wireshark, JWT Auth, RBAC, OWASP, Monitoring

Projects

SecRAG: Retrieval Augmented Generation Backend | FastAPI, ChromaDB, Sentence Transformers, React, NumPy, Python, Docker

- Built an end-to-end RAG pipeline with hybrid BM25 and dense vector retrieval fused using Reciprocal Rank Fusion, achieving 4.15/5.0 average answer quality on a 48-question golden eval dataset.
- Implemented a two-stage retrieval pipeline combining ChromaDB vector search with a cross encoder reranker to cut top 20 to top 5, improving retrieval precision over single method baseline.
- Added LLM as a judge citation verification that parses inline chunk reference and validates each claim against source text, achieving 99.6% citation accuracy across the eval suite.
- Migrate from flat file embedding storage to ChromaDB with near duplicate deduplication at ingestion, supporting three configurable chunking strategies including semantic topic boundary detection.
- Containerized the full stack with Docker Compose and built a React frontend with real time confidence scoring, retrieval mode switching, and citation display.

PulseScore: Real Time AI Inference and Monitoring System | FastAPI, Python, ML Deployment, Docker

- Built a real time AI inference backend serving ML model predictions through REST APIs with average end to end latency under 120ms, supporting sustained request rates of 100+ predictions per minute.
- Integrated trained ML models into production style backend services, managing a versioned deployment pipeline across 3 model versions to support safe rollback and controlled updates.
- Designed end to end logging and monitoring pipelines tracking prediction latency, error rates, and system health, cutting average debug time by roughly 25% through structured observability.
- Optimized API throughput by profiling request handling bottleneck, improving peak concurrency capacity by approximately 30% without additional infrastructure.

Certifications

Generative AI and LLMs | DeepLearning.AI

February 2026

Machine Learning Specialization | DeepLearning.AI

January 2026

AWS Certified Cloud Practitioner | Amazon Web Service

July 2025

AWS Academy Cloud Architect and Solutions Engineer | Amazon Web Services

April 2025